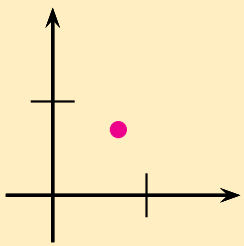
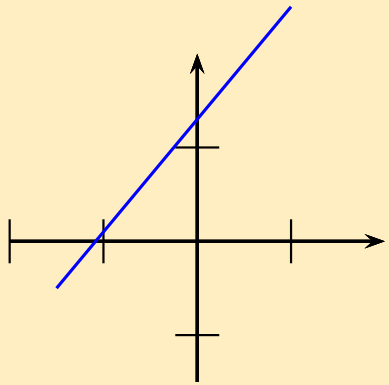
What does dimension really mean ?

Intuition

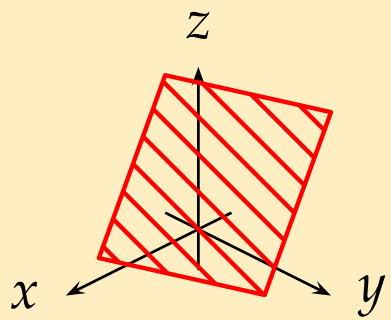
- A point is zero dimensional.
- A line is one dimensional.
- A plane is two dimensional.



zero dimensional



one dimensional



two dimensional

- Dimension means...
  - Number of independent parameters.
  - Number of independent directions of movement.
- We will obtain a more precise definition of dimension.

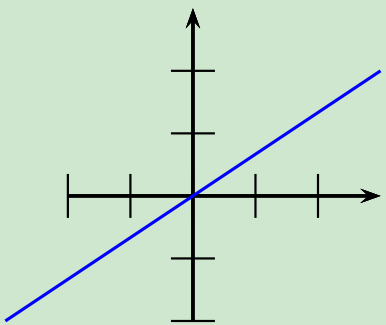
Subspaces of $\mathbb{R}^n$

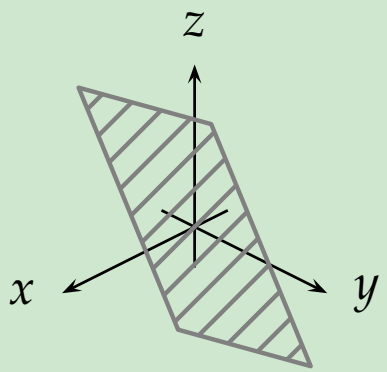
A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

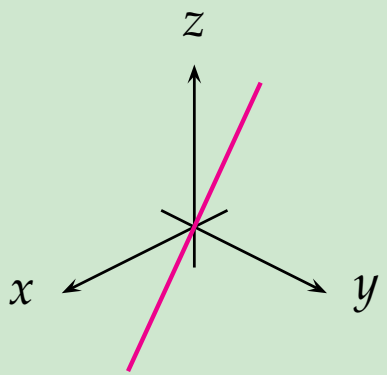
1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Example

- Every line in $\mathbb{R}^2$ or $\mathbb{R}^3$ that passes through the origin is a subspace of $\mathbb{R}^n$.
- Every plane in $\mathbb{R}^3$ that passes through the origin is a subspace of $\mathbb{R}^n$.







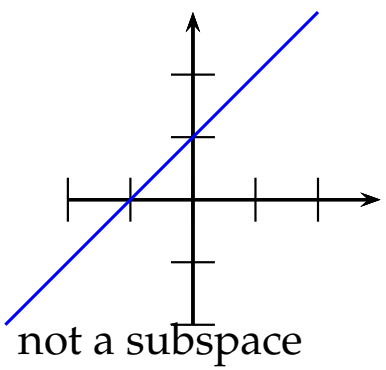
Subspaces of $\mathbb{R}^n$

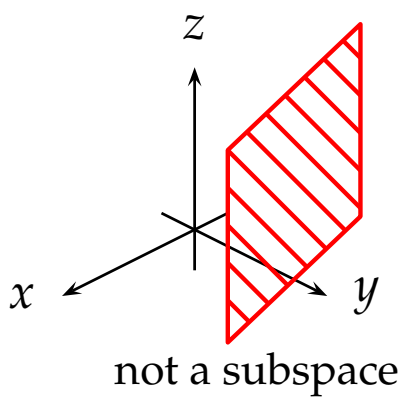
A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

1. The zero vector $\mathbf{0}$ is in H.
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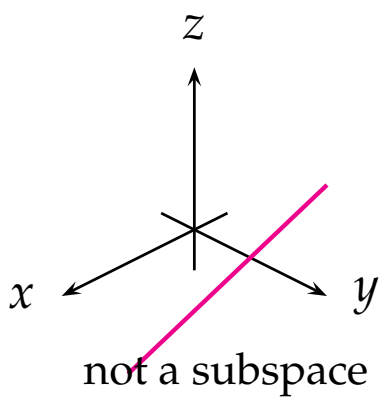
However, a line or a plane that does not pass through the origin is not a subspace.

Reason : It does not contain the zero vector. (And in fact the two other conditions fail as well.)


not a subspace



not a subspace



not a subspace

A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Example. If $\mathbf{a} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ -3 \end{bmatrix}$ then Span$\{\mathbf{a}\}$ is the set of all vectors of the form $t\mathbf{a} = \begin{bmatrix} t \\ -t \\ 2t \\ -3t \end{bmatrix}$ where t is a scalar.

1. If $t = 0$ then $0 \cdot \mathbf{a} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, i.e., the zero vector is in Span$\{\mathbf{a}\}$.
2. If $\mathbf{u}$ and $\mathbf{v}$ are vectors in H then $\mathbf{u} = s\mathbf{a}$ and $\mathbf{v} = t\mathbf{a}$. Therefore $\mathbf{u} + \mathbf{v} = s\mathbf{a} + t\mathbf{a} = (s + t)\mathbf{a}$. Therefore $\mathbf{u} + \mathbf{v}$ is also in Span$\{\mathbf{a}\}$.

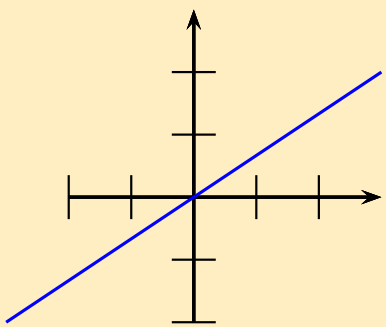
$$\begin{bmatrix} s \\ -s \\ 2s \\ -3s \end{bmatrix} + \begin{bmatrix} t \\ -t \\ 2t \\ -3t \end{bmatrix} = \begin{bmatrix} s+t \\ -(s+t) \\ 2(s+t) \\ -3(s+t) \end{bmatrix}$$

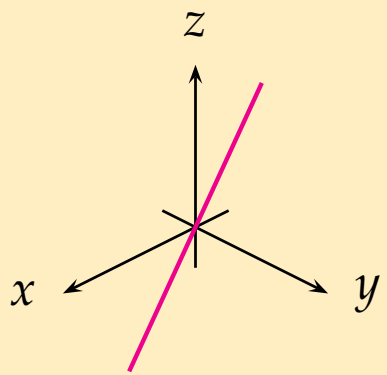
3. $\mathbf{u}$ a vector in Span$\{\mathbf{a}\}$ and c a scalar $\Rightarrow$ $\mathbf{u} = t\mathbf{a}$ for some scalar t $\Rightarrow$ $c\mathbf{u} = c(t\mathbf{a}) = (ct)\mathbf{a}$ is also in Span$\{\mathbf{a}\}$.

A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

If $\mathbf{v}$ is a vector in $\mathbb{R}^n$ then Span$\{\mathbf{v}\}$ is a subspace of $\mathbb{R}^n$.





A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Example. Is the set $H = \left\{ \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 2 \end{bmatrix} \right\}$ a subspace of $\mathbb{R}^3$?

Solution. We should check all of the above three conditions.

- If all of them hold, then H is a subspace.
- If at least one of them fails to be true, then H is not a subspace.

- The zero vector belongs to H. ✓
- If $\mathbf{u}$ and $\mathbf{v}$ belong to H then $\mathbf{u} + \mathbf{v}$ also belongs to H. ×

  Choose $\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 4 \\ -2 \\ 2 \end{bmatrix}$. Then $\mathbf{u} + \mathbf{v} = \begin{bmatrix} 5 \\ -4 \\ 5 \end{bmatrix}$ is not in H
- If $\mathbf{u}$ is in H and c is a scalar then $c\mathbf{u}$ is in H ×

  If we take $\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and $c = 3$ then $c\mathbf{u} = 3\mathbf{u} = \begin{bmatrix} 3 \\ -6 \\ 9 \end{bmatrix}$ is not in H.

A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Example

If $H = \{\mathbf{0}\} = \left\{ \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \right\}$ then H is a subspace.

However, if $H = \{\, \mathbf{a} \,\}$ and $\mathbf{a} \neq \mathbf{0}$ then H is never a subspace.

Reason: For example, if $\mathbf{a} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ then $2\mathbf{a} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$ is not in H.

A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Example. Which of the following is a subspace of $\mathbb{R}^n$ for the given n?

1. $H = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$ for $n = 2$ No!
2. $H = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$ for $n = 3$ Yes!
3. $H = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ for $n = 3$ No!
4. $H = \left\{ \begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix} \middle| \, b \text{ is in } \mathbb{R} \right\}$ for $n = 3$ Yes!

$$\begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix} = b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ and therefore } H = \text{Span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Example. Which of the following is a subspace of $\mathbb{R}^n$ for the given n?

5. $H = \left\{ \begin{bmatrix} 0 \\ t \\ t^2 \end{bmatrix} \middle| \, t \text{ is in } \mathbb{R} \right\}$ for $n = 3$ No!

   Because $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ is in H but $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$ and the latter vector is not in H.

6. $H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \middle| \, 2x - 3y - z = 0 \right\}$ for $n = 3$ Yes!

   This is a plane in $\mathbb{R}^3$ which passes through the origin. (We will come back to this example.)

A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Example. Which of the following is a subspace of $\mathbb{R}^n$ for the given n?

7. $H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; x - 3y + z = 1 \right\}$ for $n = 3$ No!

   This is a plane in $\mathbb{R}^3$ which does not pass through the origin.

8. $H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; x \geq 0 \right\}$ for $n = 3$ No!

   For example, if $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ and $c = -1$ then $c\mathbf{u} = \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix}$ and the latter vector is not in H.

Example

If $\mathbf{a}_1, .., \mathbf{a}_m$ are vectors in $\mathbb{R}^n$ then Span$\{ \mathbf{a}_1, ..., \mathbf{a}_m \}$ is a subspace of $\mathbb{R}^n$.

Reason:

- The zero vector "$\mathbf{0}$" is in Span$\{ \mathbf{a}_1, ..., \mathbf{a}_m \}$ because

$$0\mathbf{a}_1 + 0\mathbf{a}_2 + \cdots + 0\mathbf{a}_m = \mathbf{0}$$

- If $\mathbf{u} = c_1\mathbf{a}_1 + \cdots + c_m\mathbf{a}_m$ and $\mathbf{v} = d_1\mathbf{a}_1 + \cdots + d_m\mathbf{a}_m$ are in Span$\{ \mathbf{a}_1, ..., \mathbf{a}_m \}$ then

$$\mathbf{u} + \mathbf{v} = c_1\mathbf{a}_1 + \cdots + c_m\mathbf{a}_m + d_1\mathbf{a}_1 + \cdots + d_m\mathbf{a}_m = (c_1 + d_1)\mathbf{a}_1 + \cdots + (c_m + d_m)\mathbf{a}_m$$

Therefore $\mathbf{u} + \mathbf{v}$ is also in Span$\{ \mathbf{a}_1, ..., \mathbf{a}_m \}$.

- If $\mathbf{u} = c_1\mathbf{a}_1 + \cdots + c_m\mathbf{a}_m$ and c is a scalar then

$$c\mathbf{u} = c(c_1\mathbf{a}_1 + \cdots + c_m\mathbf{a}_m) = (cc_1)\mathbf{a}_1 + \cdots + (cc_m)\mathbf{a}_m$$

therefore $c\mathbf{u}$ is also in Span$\{ \mathbf{a}_1, ..., \mathbf{a}_m \}$

Example

Is $H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \,\middle|\, 2x - 3y - z = 0 \right\}$ a subspace of $\mathbb{R}^3$?

Solution.

$2x - 3y - z = 0 \;\Rightarrow\; x = \frac{3}{2}y + \frac{1}{2}z$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{3}{2}y + \frac{1}{2}z \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{3}{2}y \\ y \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{2}z \\ 0 \\ z \end{bmatrix} = y \begin{bmatrix} \frac{3}{2} \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} \frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$$

y, z free variable.

Therefore $H = \text{Span}\left\{ \begin{bmatrix} \frac{3}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ 0 \\ 1 \end{bmatrix} \right\}$, i.e., H is a subspace of $\mathbb{R}^3$.

Example

Is $H = \left\{ \begin{bmatrix} x-2y \\ y+z \\ z-x \end{bmatrix} \;\middle|\; x, y, z \in \mathbb{R} \right\}$ a subspace of $\mathbb{R}^3$?

Solution. Each vector in H can be written as:

$$\begin{bmatrix} x-2y \\ y+z \\ z-x \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ -x \end{bmatrix} + \begin{bmatrix} -2y \\ y \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ z \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + y \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Therefore $H = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$.

It follows that H is a subspace of $\mathbb{R}^3$.

Column space of a matrix

Suppose A is an $m \times n$ matrix.

The column space of A, denoted by $\operatorname{Col} A$, is the subspace of $\mathbb{R}^m$ spanned by the columns of A.

$$A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix} \Rightarrow \operatorname{Col} A = \operatorname{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$$

Example. Suppose $A = \begin{bmatrix} 1 & 3 & 1 \\ -1 & -2 & 2 \\ 0 & -1 & -3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$. Is $\mathbf{b}$ in $\operatorname{Col} A$?

Solution. This question is equivalent to the following:

$$\text{Is } \mathbf{b} \text{ in } \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \right\}?$$

And we know how to solve it! We should calculate the R.E.F. of the following matrix :

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 2 \\ -1 & -2 & 2 & 0 \\ 0 & -1 & -3 & -1 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Since the last column contains a pivot position, $\mathbf{b}$ is not in $\operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \right\}$.

Another interpretation of Col A

Suppose A is an $m \times n$ matrix.

$$A = [\ \mathbf{a}_1\ \cdots\ \mathbf{a}_n\ ]$$

Col A is the set of vectors $\mathbf{b}$ which are a linear combination of $\mathbf{a}_1, ..., \mathbf{a}_n$, i.e., $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.

This means that $\mathbf{b} = A\mathbf{x}$ where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$.

Conclusion : Col A is the set of all vectors $\mathbf{b}$ in $\mathbb{R}^m$ such that the matrix equation $A\mathbf{x} = \mathbf{b}$ is consistent.

Null space of a matrix

Suppose A is an $m \times n$ matrix.

The null space of A is the subset of $\mathbb{R}^n$ containing vectors $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$.

The null space of A is indicated by $\text{Nul}\, A$.

Theorem

$\text{Nul}\, A$ is a subspace of $\mathbb{R}^n$.

Reason.

1. $\mathbf{x} = \mathbf{0} \Rightarrow A\mathbf{x} = \mathbf{0}$. Therefore $\mathbf{0}$ is in $\text{Nul}\, A$.
2. Suppose $\mathbf{u}$ and $\mathbf{v}$ are in $\text{Nul}\, A$. Then

$$A\mathbf{u} = \mathbf{0} \text{ and } A\mathbf{v} = \mathbf{0} \quad \Longrightarrow \quad A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

   i.e., $\mathbf{u} + \mathbf{v}$ is also in $\text{Nul}\, A$.

3. Suppose $\mathbf{u}$ is in $\text{Nul}\, A$, and c is a scalar. Then $A\mathbf{u} = \mathbf{0}$ and

$$A(c\mathbf{u}) = c(A\mathbf{u}) = c\mathbf{0} = \mathbf{0}$$

   i.e., $c\mathbf{u}$ is also in $\text{Nul}\, A$.

Bases

Bases of $\mathbb{R}^n$

A list of vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ in $\mathbb{R}^n$ is called a basis of $\mathbb{R}^n$ if

1. $\text{Span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} = \mathbb{R}^n$, i.e., every vector in $\mathbb{R}^n$ is a linear combination of $\mathbf{a}_1, ..., \mathbf{a}_k$.
2. $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent.

Example. The vectors $\begin{bmatrix}1\\0\\0\end{bmatrix}$, $\begin{bmatrix}0\\1\\0\end{bmatrix}$, and $\begin{bmatrix}0\\0\\1\end{bmatrix}$ form a basis of $\mathbb{R}^3$.

1. Every vector in $\mathbb{R}^3$ is a linear combination of them :

$$\begin{bmatrix}x\\y\\z\end{bmatrix} = x\begin{bmatrix}1\\0\\0\end{bmatrix} + y\begin{bmatrix}0\\1\\0\end{bmatrix} + z\begin{bmatrix}0\\0\\1\end{bmatrix}.$$

2. Easy to check that $\begin{bmatrix}1\\0\\0\end{bmatrix}$, $\begin{bmatrix}0\\1\\0\end{bmatrix}$, and $\begin{bmatrix}0\\0\\1\end{bmatrix}$ are linearly independent.

Example

The vectors $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$ form a basis of $\mathbb{R}^2$:

1. Every vector of $\mathbf{b} = \begin{bmatrix}b_1\\b_2\end{bmatrix}$ is a linear combination of $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$:

$$\left[\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & 1 & b_2 \end{array}\right]$$

   By "Guaranteed Existence of Solution : since every row of $A = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$ has a pivot position.

2. The vectors $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$ are linearly independent.

   Because every column of the matrix $A = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$ has a pivot.

Bases of arbitrary subspaces

Suppose H is a subspace of $\mathbb{R}^n$. A list of vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ in $\mathbb{R}^n$ is called a basis of H if

1. $\mathbf{a}_1, ..., \mathbf{a}_k$ are in H.
2. $\text{Span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} = H$.
3. $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent.

- Therefore a basis is a *minimal* spanning set.
- Intuitively, a basis provides the maximum number of independent directions in H.

Question. Suppose we are given a subspace H of $\mathbb{R}^n$. How can we find a basis of H?

Answer. Next time.

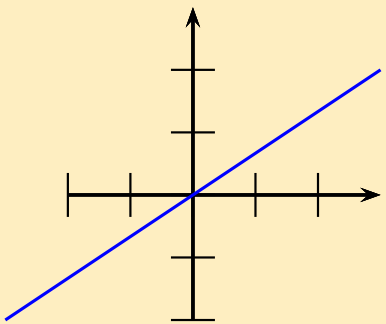
Summary

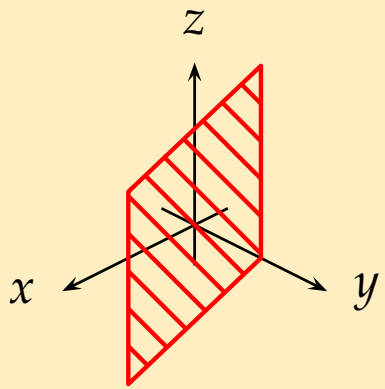
A subspace of $\mathbb{R}^n$ is any set H of vectors in $\mathbb{R}^n$ such that :

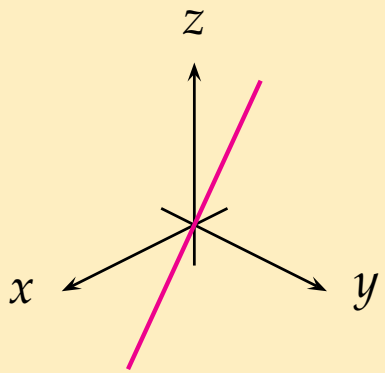
1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

If $\mathbf{a}_1, .., \mathbf{a}_m$ are vectors in $\mathbb{R}^n$ then Span$\{\mathbf{a}_1, ..., \mathbf{a}_m\}$ is a subspace of $\mathbb{R}^n$.
It follows that :

- Every line in $\mathbb{R}^n$ that passes through the origin is a subspace of $\mathbb{R}^n$.
- Every plane in $\mathbb{R}^n$ that passes through the origin is a subspace of $\mathbb{R}^n$.



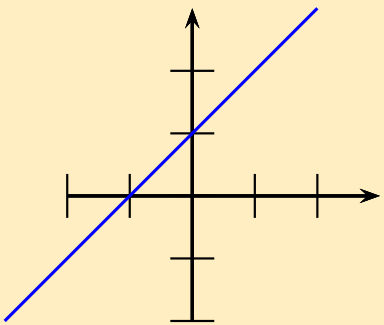




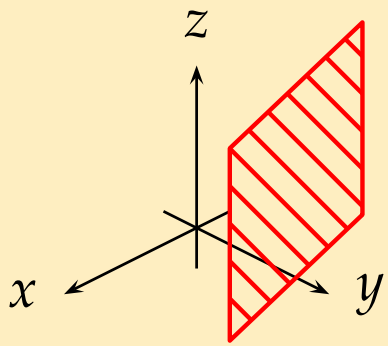
Summary

However, a line or a plane that does not pass through the origin is not a subspace.

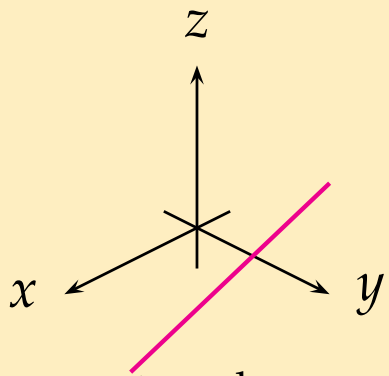
Reason : It does not contain the zero vector. (And in fact the two other conditions fail as well.)



not a subspace



not a subspace



not a subspace